

Lab 2 Report Kevin Lin

Section 1: Required Design Files

See Appendix for RTL, testbench, and SDC files

Section 2: Xcelium Simulation Results

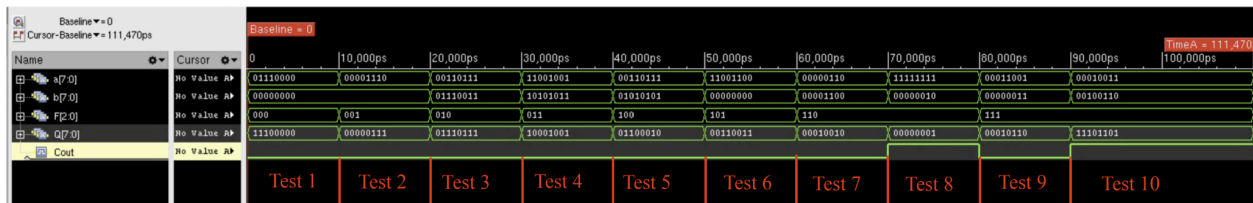


Figure 1: Outputs of lab2_alu_test.v tests using Xcelium Simulation

Table 1: Input values, expected output and their output for ALU validation

Test Case	A Value	B Value	F (mode)	Expected Out	Actual Out
1: LSL	01110000	—	000 (LSL)	11100000	11100000
2: LSR	00001110	—	001 (LSR)	00000111	00000111
3: OR	00110111	01110011	010 (OR)	01110111	01110111
4: AND	11001001	10101011	011 (AND)	10001001	10001001
5: XOR	00110111	01010101	100 (XOR)	01100010	01100010
6: NOT	11001100	—	101 (NOT)	00110011	00110011
7: ADD (no carry)	00000110 (d: 6)	00001100 (d: 12)	110 (ADD)	00010010 (d: 18)	00010010
8: ADD (w/ carry)	11111111 (d: 255)	00000010 (d: 2)	110 (ADD)	00000001 (d: 1)	00000001
9: SUB (no borrow)	00011001 (d: 25)	00000011 (d: 3)	111 (SUB)	00010110 (d: 22)	00010110
10: SUB (w/ borrow)	00010011 (d: 19)	00100110 (d: 38)	111 (SUB)	11101101 (d: -19, 2's C)	11101101

From the Xcelium results in table 1, all test cases for each of the operations work as intended. Additionally, both the ADD and SUB operations were tested twice to see if the carry/borrow bit was high when a carry/borrow occurred in the operation. For tests 8 and 10, the Cout pin was high, indicating that the carry out pin works as intended.

Section 3: Synthesis

Part A: Low Effort

Table 1: Screenshots of area, cell counts, timing and power reports from low effort synthesis

Report type	Screenshot
Area/Cell Counts	<pre> Instance Module Cell Count Cell Area Net Area Total Area ----- lab2_alu 145 187.264 210.359 397.623 </pre>
Power	<pre> Leakage Dynamic Total Instance Cells Power(nW) Power(nW) Power(nW) ----- lab2_alu 145 4113.258 22445.196 26558.454 </pre>
Timing	<pre> Path 1: MET (0 ps) Path Delay Check Startpoint: (F) b[1] Endpoint: (F) Q[7] Capture Launch Path Delay:++ 500 - Drv Adjust:++ 0 0 Arrival:= 500 Required Time:= 500 Data Path:- 500 Slack:= 0 Exceptions/Constraints: max_delay 500 lab2_alu.sdc_line_15 #----- # Timing Point Flags Arc Edge Cell Fanout Load Trans Delay Arrival Instance # (fF) (ps) (ps) (ps) (ps) (ps) Location #----- b[1] - - F (arrival) 2 4.6 0 0 0 (-,-) fopt1614/Z - - F CLKBUF_X3 4 7.8 10 31 31 (-,-) add_36_20_g193_5953/ZN - A->Z F NAND2_X1 1 3.8 14 22 53 (-,-) add_36_20_g182_2683/ZN - A2->ZN F NAND2_X2 2 4.2 8 16 69 (-,-) add_36_20_g181_9682/ZN - A1->ZN R NAND2_X1 1 1.9 10 15 84 (-,-) add_36_20_g179_1474/ZN - A1->ZN F NAND2_X1 1 3.0 9 16 99 (-,-) add_36_20_g178_3772/CO - CI->CO F FA_X1 1 3.0 15 70 169 (-,-) add_36_20_g177_4296/CO - CI->CO F FA_X1 1 3.0 15 72 242 (-,-) add_36_20_g176_8780/CO - CI->CO F FA_X1 1 3.0 15 72 314 (-,-) add_36_20_g175_9906/CO - CI->CO F FA_X1 1 2.4 14 71 384 (-,-) add_36_20_g174_1857/ZN - A->ZN F XNOR2_X1 1 3.5 15 44 428 (-,-) g1297_3772/ZN - B1->ZN R AOI221_X2 1 2.0 37 66 495 (-,-) g1295/ZN - A->ZN F INV_X1 1 0.4 8 5 500 (-,-) Q[7] <<< - F (port) - - - 0 500 (-,-) #----- </pre>

Part B: High Effort

Table 2: Screenshots of area, cell counts, timing and power reports from high effort synthesis

Report type	Screenshot
Area/Cell Counts	<pre> Instance Module Cell Count Cell Area Net Area Total Area ----- lab2_alu 140 142.842 206.122 348.964 </pre>
Power	<pre> Leakage Dynamic Total Instance Cells Power(nW) Power(nW) Power(nW) ----- lab2_alu 140 3512.676 17226.602 20739.279 </pre>
Timing	<pre> Path 1: MET (40 ps) Path Delay Check Startpoint: (F) a[1] Endpoint: (R) Q[7] Capture Launch Path Delay:+ 500 - Drv Adjust:+ 0 0 Arrival:= 500 Required Time:= 500 Data Path:- 460 Slack:= 40 Exceptions/Constraints: max_delay 500 lab2_alu.sdc_line_15 #----- # Timing Point Flags Arc Edge Cell Fanout Load Trans Delay Arrival Instance # (ps) (ps) (ps) (ps) (ps) Location #----- a[1] - - F (arrival) 1 1.0 0 0 0 (-,-) drc_buf_sp/Z - A->Z F CLKBUF_X1 6 9.6 24 47 47 (-,-) g3778_7114/ZN - A1->ZN R NOR2_X1 3 5.7 34 50 98 (-,-) g3725_5703/ZN - A1->ZN F NOR2_X1 1 1.7 10 11 109 (-,-) g3724_1474/ZN - A1->ZN R NOR2_X1 2 3.9 26 35 144 (-,-) g3702_7114/ZN - B1->ZN F OAI21_X1 2 2.8 14 21 164 (-,-) g3686_9906/ZN - B1->ZN R AOI21_X1 2 3.9 32 39 203 (-,-) g3679_7344/ZN - B1->ZN F OAI21_X1 2 2.8 14 22 225 (-,-) g3673_5953/ZN - B1->ZN R AOI21_X1 2 3.9 32 39 264 (-,-) g3665_6877/ZN - B1->ZN F OAI21_X1 1 1.2 11 17 281 (-,-) g3663_2683/Z - A->Z F MUX2_X1 1 2.4 11 61 342 (-,-) g3660_1474/ZN - A->ZN F XNOR2_X1 1 1.2 13 37 379 (-,-) g3657_4296/Z - B->Z F MUX2_X1 1 1.8 10 60 440 (-,-) g3656_8780/ZN - B1->ZN R OAI21_X1 1 0.4 13 20 460 (-,-) Q[7] <<< - R (port) - - - 0 460 (-,-) #----- </pre>

From the data in these two timing reports, all 4 parameters (area, cell count, timing and power) were improved after using Synthesis with high effort compared to Synthesis with low effort.

Section 4: Generated Gate level netlist

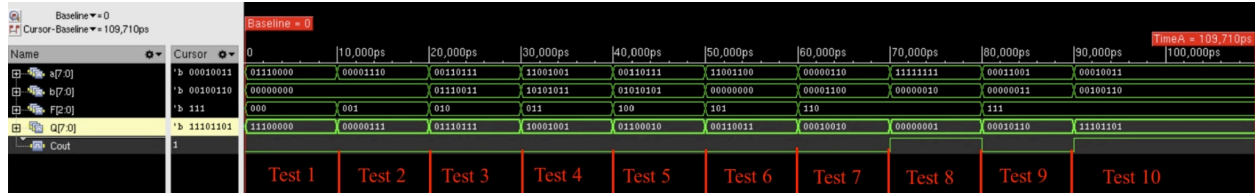


Figure 2: Outputs of lab2_alu_test.v using Generated Gate Level Netlist

Table 1: Input values, expected output and their output for ALU validation

Test Case	A Value	B Value	F (mode)	Expected Out	Actual Out
1: LSL	01110000	–	000 (LSL)	11100000	11100000
2: LSR	00001110	–	001 (LSR)	00000111	00000111
3: OR	00110111	01110011	010 (OR)	01110111	01110111
4: AND	11001001	10101011	011 (AND)	10001001	10001001
5: XOR	00110111	01010101	100 (XOR)	01100010	01100010
6: NOT	11001100	–	101 (NOT)	00110011	00110011
7: ADD (no carry)	00000110 (d: 6)	00001100 (d: 12)	110 (ADD)	00010010 (d: 18)	00010010
8: ADD (w/ carry)	11111111 (d: 255)	00000010 (d: 2)	110 (ADD)	00000001 (d: 1)	00000001
9: SUB (no borrow)	00011001 (d: 25)	00000011 (d: 3)	111 (SUB)	00010110 (d: 22)	00010110
10: SUB (w/ borrow)	00010011 (d: 19)	00100110 (d: 38)	111 (SUB)	11101101 (d: -19, 2's C)	11101101

After running the same tests on the gate level netlist, we find that the results are identical to the Xcelium simulation results.

Section 5: Appendix

File 1: RTL Code – lab2_alu.v

```
`timescale 1ns/10ps

module lab2_alu(a, b, F, Q, Cout);
    input [7:0] a, b;
    input [2:0] F;
    output [7:0] Q;
    output Cout;

    // internal wires for Q computation
    wire [7:0] lsl_out;
    wire [7:0] lsr_out;
    wire [7:0] or_out;
    wire [7:0] and_out;
    wire [7:0] xor_out;
    wire [7:0] not_out;

    wire [8:0] add;
    wire [8:0] sub;
    wire [7:0] add_out;
    wire [7:0] sub_out;

    // wires for Cout computation
    wire carry_out;
    wire borrow_out;

    // Computing all Q values
    assign lsl_out = a << 1;
    assign lsr_out = a >> 1;
    assign or_out = a | b;
    assign and_out = a & b;
    assign xor_out = a ^ b;
    assign not_out = ~a;

    // Including 9th bit for Cout computation
    assign add = a + b;
    assign sub = a - b;
```

```

assign add_out = add[7:0];
assign sub_out = sub[7:0];

assign carry_out = add[8];
assign borrow_out = ~sub[8];

assign Q = (F == 0)? lsl_out:
  ((F == 1)? lsr_out:
  ((F == 2)? or_out:
  ((F == 3)? and_out:
  ((F == 4)? xor_out:
  ((F == 5)? not_out:
  ((F == 6)? add_out:
  sub_out)))));

assign Cout = (F == 6)? add_out:
  ((F == 7)? sub_out: 0);
endmodule

```

File 2: Testbench – lab2_alu_test.v

```

`timescale 1ns/10ps

module lab2_alu_test;
  reg [7:0] a, b;
  reg [2:0] F;
  wire [7:0] Q;
  wire Cout;

  // Creating instance of lab2_alu module
  lab2_alu test_alu(.a(a), .b(b), .F(F), .Q(Q), .Cout(Cout));

  initial begin
    // lsl test
    a = 8'b01110000;
    b = 8'h00;
    F = 3'b000;
    #10
    // lsr test

```

```
a = 8'b00001110;
b = 8'h00;
F = 3'b001;
#10
// or test
a = 8'b00110111;
b = 8'b01110011;
F = 3'b010;
#10
// and test
a = 8'b11001001;
b = 8'b10101011;
F = 3'b011;
#10
// xor test
a = 8'b00110111;
b = 8'b01010101;
F = 3'b100;
#10
// not test
a = 8'b11001100;
b = 8'h00;
F = 3'b101;
#10
// add test (no Cout) 0x6 + 0xc = 0x12
a = 8'h06;
b = 8'h0c;
F = 3'b110;
#10
// add test (active Cout) 0xff + 0x2
a = 8'hff;
b = 8'h02;
F = 3'b110;
#10
// sub test (no Cout) 0x19 - 0x3 = 0x16
a = 8'h19;
b = 8'h03;
F = 3'b111;
#10
// sub test (active Cout) 0x13 - 0x26
```

```

    a = 8'h13;
    b = 8'h26;
    F = 3'b111;

    #20 $finish;
end
endmodule

```

File 3: SDC file – lab2_alu.sdc

```

#create_clock -name clk -period 1.0 -waveform { 0 0.5 } [get_ports clk]

# ----- Input constraints -----

#set_input_delay -clock clk -max 0.2 [get_ports {din start rstb wr_ctrl_test_ctrl}]
#set_input_delay -clock clk -min -0.2 [get_ports {din start rstb wr_ctrl_test_ctrl}]

# ----- Output constraints -----

#set_output_delay -clock clk -max 0.2 [get_ports {addr_out*}]
#set_output_delay -clock clk -min -0.2 [get_ports {addr_out*}]

# Changed max delay to 0.5
set_max_delay 0.5 -from [all_inputs] -to [all_outputs]

# Assume 50fF load capacitances everywhere:
set_load 0.050 [all_outputs]
# Set 10fF maximum capacitance on all inputs
set_max_capacitance 0.010 [all_inputs]

# set clock uncertainty of the system clock (skew and jitter)
#set_clock_uncertainty -setup 0.03 [get_clocks clk*]
#set_clock_uncertainty -hold 0.06 [get_clocks clk*]

# set maximum transition at output ports
set_max_transition 0.07 [current_design]

# set_attr use_scan_seqs_for_non_dft false

```

File 4: Synthesis Script – synth.tcl

1. Setup Libraries

```
set_db library /vol/ece303/genus_tutorial/NangateOpenCellLibrary_typical.lib  
set_db lef_library /vol/ece303/genus_tutorial/NangateOpenCellLibrary.lef
```

2. Read and Elaborate Design

```
read_hdl ../lab2_alu.v  
elaborate  
current_design lab2_alu
```

3. Apply Constraints

```
read_sdc ../lab2_alu.sdc
```

4. Synthesize

```
syn_generic -effort high  
syn_map  
syn_opt
```

5. Generate Reports

```
report_timing > timing.rpt  
report_area > area.rpt  
report_power > power.rpt
```

6. Export Output

```
write_hdl > lab2_alu_syn.v
```

File 5: Synthesis – lab2_alu_syn.v

```
`timescale 1ns/10ps

// Generated by Cadence Genus(TM) Synthesis Solution 18.14-s037_1
// Generated on: Apr 24 2026 17:35:51 CDT (Apr 24 2026 22:35:51 UTC)

// Verification Directory fv/lab2_alu

module lab2_alu(a, b, F, Q, Cout);
  input [7:0] a, b;
  input [2:0] F;
  output [7:0] Q;
  output Cout;
  wire [7:0] a, b;
  wire [2:0] F;
  wire [7:0] Q;
  wire Cout;
  wire n_0, n_1, n_2, n_3, n_4, n_5, n_6, n_7;
  wire n_8, n_9, n_10, n_11, n_12, n_13, n_14, n_15;
  wire n_16, n_17, n_18, n_19, n_20, n_21, n_22, n_23;
  wire n_24, n_25, n_26, n_27, n_28, n_29, n_30, n_31;
  wire n_32, n_33, n_34, n_35, n_36, n_37, n_38, n_39;
  wire n_40, n_41, n_42, n_43, n_44, n_45, n_46, n_47;
  wire n_48, n_50, n_51, n_52, n_53, n_54, n_55, n_56;
  wire n_57, n_58, n_59, n_60, n_61, n_62, n_63, n_64;
  wire n_65, n_66, n_67, n_68, n_69, n_70, n_71, n_72;
  wire n_73, n_74, n_75, n_76, n_77, n_78, n_79, n_80;
  wire n_81, n_82, n_83, n_84, n_85, n_86, n_87, n_88;
  wire n_89, n_90, n_91, n_92, n_93, n_94, n_95, n_97;
  wire n_98, n_99, n_100, n_101, n_102, n_103, n_105, n_106;
  wire n_107, n_108, n_109, n_110, n_111, n_113, n_114, n_116;
  wire n_117, n_118, n_119, n_120, n_121, n_122, n_123, n_124;
  wire n_125, n_126, n_127, n_128, n_129, n_132, n_133, n_134;
  wire n_135, n_136, n_137, n_139;
  OAI221_X1 g3716__8780(.A (n_139), .B1 (n_137), .B2 (n_43), .C1
    (n_45), .C2 (n_44), .ZN (Q[7]));
  AOI211_X1 g3717__4296(.A (n_72), .B (n_136), .C1 (n_90), .C2 (a[6]),
    .ZN (n_139));
  OR2_X1 g3720__3772(.A1 (n_133), .A2 (n_134), .ZN (Q[6]));
```

AOI21_X1 g3718__1474(.A (n_128), .B1 (n_135), .B2 (n_127), .ZN (n_137));
NOR3_X1 g3719__4547(.A1 (n_135), .A2 (n_42), .A3 (n_132), .ZN (n_136));
OAI221_X1 g3723__9682(.A (n_59), .B1 (n_129), .B2 (n_125), .C1 (n_95), .C2 (a[6]), .ZN (n_134));
OAI221_X1 g3722__2683(.A (n_55), .B1 (n_126), .B2 (n_132), .C1 (n_78), .C2 (n_0), .ZN (n_133));
OAI211_X1 g3724__1309(.A (n_124), .B (n_91), .C1 (n_120), .C2 (n_32), .ZN (Q[5]));
AOI222_X1 g3721__6877(.A1 (n_116), .A2 (n_74), .B1 (n_117), .B2 (a[6]), .C1 (n_123), .C2 (b[6]), .ZN (n_135));
NAND4_X1 g3728__2900(.A1 (n_119), .A2 (n_107), .A3 (n_58), .A4 (n_50), .ZN (Q[4]));
AOI21_X1 g3725__2391(.A (n_128), .B1 (n_122), .B2 (n_127), .ZN (n_129));
NAND2_X1 g3727__7675(.A1 (n_121), .A2 (n_125), .ZN (n_126));
AOI22_X1 g3729__7118(.A1 (n_114), .A2 (n_127), .B1 (n_118), .B2 (n_4), .ZN (n_124));
NAND2_X1 g3726__8757(.A1 (n_122), .A2 (n_20), .ZN (n_123));
INV_X1 g3730(.A (n_122), .ZN (n_121));
AOI21_X1 g3732__1786(.A (n_128), .B1 (n_113), .B2 (n_127), .ZN (n_120));
AOI222_X1 g3733__5953(.A1 (n_101), .A2 (n_97), .B1 (n_118), .B2 (n_99), .C1 (n_111), .C2 (n_46), .ZN (n_119));
NOR2_X1 g3731__5703(.A1 (n_117), .A2 (n_116), .ZN (n_122));
OAI211_X1 g3735__7114(.A (n_105), .B (n_89), .C1 (n_103), .C2 (n_26), .ZN (Q[3]));
NOR2_X1 g3734__5266(.A1 (n_113), .A2 (n_31), .ZN (n_114));
NAND4_X1 g3741__2250(.A1 (n_102), .A2 (n_87), .A3 (n_54), .A4 (n_51), .ZN (Q[2]));
AOI211_X1 g3736__6083(.A (F[0]), .B (n_75), .C1 (n_109), .C2 (n_3), .ZN (n_117));
OAI21_X1 g3738__2703(.A (n_84), .B1 (n_106), .B2 (n_132), .ZN (n_111));
AOI211_X1 g3737__5795(.A (n_108), .B (n_13), .C1 (n_110), .C2 (n_10), .ZN (n_116));
MUX2_X1 g3740__7344(.A (n_110), .B (n_109), .S (n_108), .Z (n_113));
NAND3_X1 g3739__1840(.A1 (n_106), .A2 (n_47), .A3 (n_127), .ZN (n_107));

AOI22_X1 g3742__5019(.A1 (n_94), .A2 (n_127), .B1 (n_118), .B2 (n_16), .ZN (n_105));

OAI211_X1 g3749__1857(.A (n_92), .B (n_67), .C1 (n_73), .C2 (n_28), .ZN (Q[1]));

AOI21_X1 g3743__9906(.A (n_128), .B1 (n_93), .B2 (n_127), .ZN (n_103));

OAI21_X1 g3744__8780(.A (n_35), .B1 (n_100), .B2 (n_24), .ZN (n_110));

AOI222_X1 g3745__4296(.A1 (n_101), .A2 (n_60), .B1 (n_118), .B2 (n_62), .C1 (n_85), .C2 (n_86), .ZN (n_102));

MUX2_X1 g3747__3772(.A (n_100), .B (n_98), .S (n_108), .Z (n_106));

AOI21_X1 g3746__1474(.A (n_99), .B1 (n_98), .B2 (n_97), .ZN (n_109));

OAI211_X1 g3750__4547(.A (n_81), .B (n_65), .C1 (n_95), .C2 (a[0]), .ZN (Q[0]));

NOR2_X1 g3748__9682(.A1 (n_93), .A2 (n_25), .ZN (n_94));

AOI221_X1 g3753__2683(.A (n_66), .B1 (n_69), .B2 (n_127), .C1 (n_64), .C2 (a[2]), .ZN (n_92));

AOI221_X1 g3762__1309(.A (n_77), .B1 (n_90), .B2 (a[4]), .C1 (n_88), .C2 (n_12), .ZN (n_91));

AOI221_X1 g3763__6877(.A (n_80), .B1 (n_90), .B2 (a[2]), .C1 (n_88), .C2 (n_17), .ZN (n_89));

OAI21_X1 g3751__2900(.A (n_15), .B1 (n_82), .B2 (n_79), .ZN (n_98));

AOI21_X1 g3752__2391(.A (n_7), .B1 (n_83), .B2 (n_18), .ZN (n_100));

OR3_X1 g3754__7675(.A1 (n_71), .A2 (n_86), .A3 (n_132), .ZN (n_87));

OAI21_X1 g3755__7118(.A (n_84), .B1 (n_70), .B2 (n_132), .ZN (n_85));

MUX2_X1 g3756__8757(.A (n_83), .B (n_82), .S (n_108), .Z (n_93));

AOI22_X1 g3764__1786(.A1 (n_53), .A2 (n_48), .B1 (n_118), .B2 (n_52), .ZN (n_81));

OAI22_X1 g3774__5953(.A1 (n_76), .A2 (n_79), .B1 (n_78), .B2 (n_57), .ZN (n_80));

OAI22_X1 g3776__5703(.A1 (n_76), .A2 (n_75), .B1 (n_78), .B2 (n_74), .ZN (n_77));

AOI21_X1 g3757__7114(.A (n_128), .B1 (n_68), .B2 (n_127), .ZN (n_73));

OAI21_X1 g3758__5266(.A (n_56), .B1 (n_95), .B2 (a[7]), .ZN (n_72));

INV_X1 g3761(.A (n_70), .ZN (n_71));

NOR2_X1 g3768__2250(.A1 (n_68), .A2 (n_27), .ZN (n_69));

AOI22_X1 g3773__6083(.A1 (n_101), .A2 (n_38), .B1 (n_118), .B2 (n_39), .ZN (n_67));

MUX2_X1 g3765__2703(.A (n_63), .B (n_61), .S (n_108), .Z (n_70));

OAI22_X1 g3770__5795(.A1 (n_41), .A2 (n_11), .B1 (n_95), .B2 (a[1]), .ZN (n_66));

AOI22_X1 g3775__7344(.A1 (n_101), .A2 (n_6), .B1 (n_64), .B2 (a[1]),
.ZN (n_65));
OAI21_X1 g3759__1840(.A (n_36), .B1 (n_63), .B2 (n_22), .ZN (n_83));
AOI21_X1 g3760__5019(.A (n_62), .B1 (n_61), .B2 (n_60), .ZN (n_82));
OAI21_X1 g3766__1857(.A (n_101), .B1 (a[6]), .B2 (b[6]), .ZN (n_59));
AOI22_X1 g3771__9906(.A1 (n_90), .A2 (a[3]), .B1 (n_88), .B2 (n_57),
.ZN (n_58));
OAI21_X1 g3767__8780(.A (n_101), .B1 (a[7]), .B2 (b[7]), .ZN (n_56));
AOI22_X1 g3769__4296(.A1 (n_90), .A2 (a[5]), .B1 (n_37), .B2 (a[6]),
.ZN (n_55));
AOI22_X1 g3772__3772(.A1 (n_90), .A2 (a[1]), .B1 (n_88), .B2 (n_2),
.ZN (n_54));
NAND2_X1 g3782__1474(.A1 (n_84), .A2 (n_132), .ZN (n_53));
AOI22_X1 g3787__4547(.A1 (n_30), .A2 (F[0]), .B1 (n_52), .B2 (n_108),
.ZN (n_68));
INV_X1 g3803(.A (n_101), .ZN (n_76));
NAND2_X1 g3781__9682(.A1 (n_64), .A2 (a[3]), .ZN (n_51));
NAND2_X1 g3780__2683(.A1 (n_64), .A2 (a[5]), .ZN (n_50));
AND2_X1 g3784__1309(.A1 (n_48), .A2 (n_127), .ZN (Cout));
INV_X1 g3783(.A (n_40), .ZN (n_61));
INV_X1 g3789(.A (n_46), .ZN (n_47));
AOI21_X1 g3786__6877(.A (n_9), .B1 (n_14), .B2 (n_29), .ZN (n_63));
INV_X1 g3802(.A (n_128), .ZN (n_84));
INV_X1 g3801(.A (n_88), .ZN (n_95));
NOR2_X1 g3810__2900(.A1 (n_45), .A2 (F[0]), .ZN (n_101));
INV_X1 g3778(.A (n_33), .ZN (n_44));
INV_X1 g3777(.A (n_42), .ZN (n_43));
INV_X1 g3792(.A (n_90), .ZN (n_41));
AOI21_X1 g3785__2391(.A (n_39), .B1 (n_52), .B2 (n_38), .ZN (n_40));
AND2_X1 g3793__7675(.A1 (n_118), .A2 (b[6]), .ZN (n_37));
NAND2_X1 g3798__7118(.A1 (n_21), .A2 (n_36), .ZN (n_86));
NAND2_X1 g3795__8757(.A1 (n_23), .A2 (n_35), .ZN (n_46));
INV_X1 g3800(.A (n_64), .ZN (n_78));
NOR2_X1 g3808__1786(.A1 (n_34), .A2 (n_108), .ZN (n_88));
NOR2_X1 g3809__5953(.A1 (n_34), .A2 (F[0]), .ZN (n_128));
HA_X1 g3779__5703(.A (b[7]), .B (a[7]), .CO (n_33), .S (n_42));
INV_X1 g3788(.A (n_31), .ZN (n_32));
INV_X1 g3812(.A (n_29), .ZN (n_30));
INV_X1 g3790(.A (n_27), .ZN (n_28));
INV_X1 g3791(.A (n_25), .ZN (n_26));

INV_X1 g3814(.A (n_118), .ZN (n_45));
XOR2_X1 g3806__7114(.A (n_74), .B (b[6]), .Z (n_125));
INV_X1 g3813(.A (n_23), .ZN (n_24));
INV_X1 g3811(.A (n_21), .ZN (n_22));
XOR2_X1 g3804__5266(.A (F[0]), .B (n_74), .Z (n_20));
NOR2_X1 g3805__2250(.A1 (n_52), .A2 (n_5), .ZN (n_48));
AND2_X1 g3807__6083(.A1 (n_19), .A2 (F[0]), .ZN (n_64));
AND2_X1 g3799__2703(.A1 (n_19), .A2 (n_108), .ZN (n_90));
NAND2_X1 g3821__5795(.A1 (n_17), .A2 (b[3]), .ZN (n_18));
INV_X1 g3829(.A (n_15), .ZN (n_16));
NAND2_X1 g3819__7344(.A1 (n_8), .A2 (b[1]), .ZN (n_14));
NOR2_X1 g3824__1840(.A1 (n_12), .A2 (b[5]), .ZN (n_13));
XOR2_X1 g3796__5019(.A (a[1]), .B (b[1]), .Z (n_27));
NAND2_X1 g3817__1857(.A1 (n_11), .A2 (b[0]), .ZN (n_29));
XOR2_X1 g3794__9906(.A (a[5]), .B (b[5]), .Z (n_31));
XOR2_X1 g3797__8780(.A (a[3]), .B (b[3]), .Z (n_25));
NAND2_X1 g3815__4296(.A1 (n_1), .A2 (F[2]), .ZN (n_34));
INV_X1 g3831(.A (n_132), .ZN (n_127));
NAND2_X1 g3823__3772(.A1 (n_12), .A2 (b[5]), .ZN (n_10));
NOR2_X1 g3822__1474(.A1 (n_8), .A2 (b[1]), .ZN (n_9));
NOR2_X1 g3820__4547(.A1 (n_17), .A2 (b[3]), .ZN (n_7));
INV_X1 g3830(.A (n_5), .ZN (n_6));
INV_X1 g3828(.A (n_3), .ZN (n_4));
OR2_X1 g3825__9682(.A1 (n_2), .A2 (b[2]), .ZN (n_36));
NAND2_X1 g3826__2683(.A1 (n_57), .A2 (b[4]), .ZN (n_23));
OR2_X1 g3818__1309(.A1 (n_57), .A2 (b[4]), .ZN (n_35));
NAND2_X1 g3816__6877(.A1 (n_2), .A2 (b[2]), .ZN (n_21));
NOR2_X1 g3827__2900(.A1 (n_1), .A2 (F[2]), .ZN (n_118));
NOR2_X1 g3843__2391(.A1 (a[3]), .A2 (b[3]), .ZN (n_79));
NAND2_X1 g3839__7675(.A1 (a[5]), .A2 (b[5]), .ZN (n_3));
NOR2_X1 g3837__7118(.A1 (a[5]), .A2 (b[5]), .ZN (n_75));
OR2_X1 g3833__8757(.A1 (a[2]), .A2 (b[2]), .ZN (n_60));
AND2_X1 g3836__1786(.A1 (a[1]), .A2 (b[1]), .ZN (n_39));
NOR2_X1 g3842__5953(.A1 (a[0]), .A2 (b[0]), .ZN (n_5));
AND2_X1 g3841__5703(.A1 (a[2]), .A2 (b[2]), .ZN (n_62));
OR2_X1 g3834__7114(.A1 (a[4]), .A2 (b[4]), .ZN (n_97));
NOR2_X1 g3838__5266(.A1 (F[2]), .A2 (F[1]), .ZN (n_19));
OR2_X1 g3832__2250(.A1 (a[1]), .A2 (b[1]), .ZN (n_38));
AND2_X1 g3835__6083(.A1 (a[4]), .A2 (b[4]), .ZN (n_99));
NAND2_X1 g3840__2703(.A1 (a[3]), .A2 (b[3]), .ZN (n_15));

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AND2_X1 g3844__5795(.A1 (a[0]), .A2 (b[0]), .ZN (n_52));
NAND2_X1 g3845__7344(.A1 (F[2]), .A2 (F[1]), .ZN (n_132));
INV_X1 g3850(.A (a[1]), .ZN (n_8));
INV_X1 g3849(.A (a[5]), .ZN (n_12));
INV_X1 g3854(.A (a[3]), .ZN (n_17));
INV_X1 g3848(.A (a[4]), .ZN (n_57));
INV_X1 g3851(.A (F[0]), .ZN (n_108));
INV_X1 g3847(.A (a[7]), .ZN (n_0));
INV_X1 g3852(.A (F[1]), .ZN (n_1));
INV_X1 g3846(.A (a[0]), .ZN (n_11));
INV_X1 g3853(.A (a[2]), .ZN (n_2));
INV_X1 g3855(.A (a[6]), .ZN (n_74));
endmodule
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